

RLC

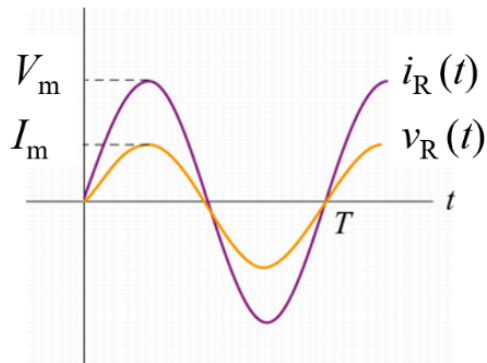
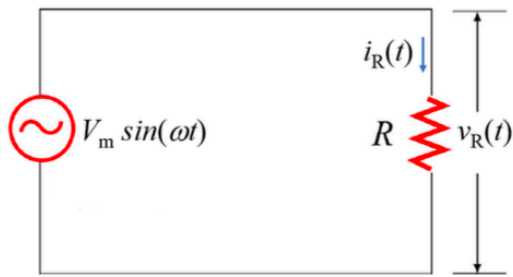
Component	Description
Inductor (L)	Stores energy in a magnetic field, resists current changes, measured in henries (H).
Capacitor (C)	Stores energy in an electric field, resists voltage changes, measured in farads (F).
Resistor (R)	Limits current flow, dissipates energy as heat, measured in ohms (Ω).
Resonant frequency (f_r)	Frequency at which a circuit exhibits maximum response.
Quality factor (Q)	Measures resonance sharpness, ratio of stored to dissipated energy.

	Resistor	Inductor	Capacitor
Current waveform	$I_m \sin(\omega t)$	$I_m \sin(\omega t - \frac{\pi}{2})$	$I_m \sin(\omega t + \frac{\pi}{2})$
Resistance/ reactance	R	$X_L = \omega L$	$X_C = \frac{1}{\omega C}$
Current amplitude	$\frac{V_m}{R}$	$\frac{V_m}{X_L}$	$\frac{V_m}{X_C}$
RMS current	$\frac{V_{rms}}{R}$	$\frac{V_{rms}}{X_L}$	$\frac{V_{rms}}{X_C}$
Phase of current	in-phase with v	lags v by $\frac{\pi}{2}$	leads v by $\frac{\pi}{2}$

	Resistor	Capacitor	Inductor
Resistance/Reactance	R	$X_C = \frac{1}{\omega C}$	$X_L = \omega L$
Phase shift	i and v are in phase	i leads v by $\frac{\pi}{2}$ rad	i lags v by $\frac{\pi}{2}$ rad
Voltage waveform	$V_m \sin(\omega t)$		
Current waveform	$\frac{V_m}{R} \sin(\omega t)$	$\frac{V_m}{X_C} \sin(\omega t + \frac{\pi}{2})$	$\frac{V_m}{X_L} \sin(\omega t - \frac{\pi}{2})$
Logic	Ohm's Law	Sinusoidal voltage divided by reactance with $\pm \frac{\pi}{2}$ rad phase difference	

Component	Equation
Resistor	$i = \frac{v}{R}$
Inductor	$v = L \frac{di}{dt}$
Capacitor	$i = C \frac{dv}{dt}$

Resistor



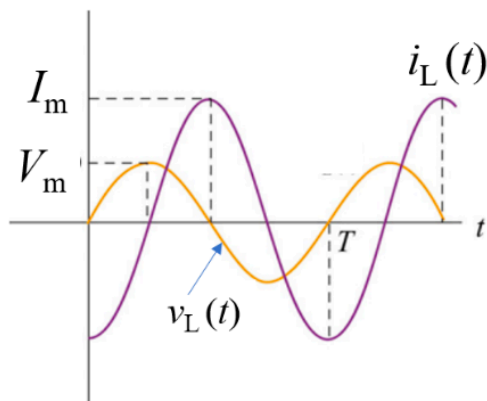
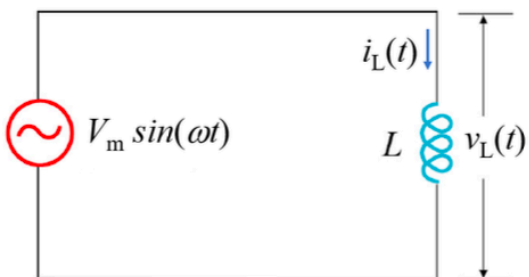
$$v_R(t) = V_m \sin(\omega t)$$

$$i_R(t) = \frac{V_m \sin(\omega t)}{R} = I_m \sin(\omega t)$$

$$I_m = \frac{V_m}{R} \quad \& \quad I_{rms} = \frac{V_{rms}}{R}$$

$\phi = 0^\circ$ (Voltage and current in-phase)

Inductor



$$v_L(t) = V_m \sin(\omega t)$$

$$v_L(t) = L \frac{di_L}{dt}$$

$$i_L = \frac{V_m}{\omega L} \sin(\omega t - \frac{\pi}{2}) = I_m \sin(\omega t - \frac{\pi}{2})$$

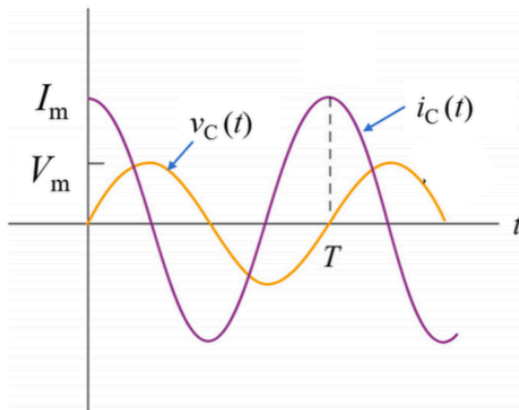
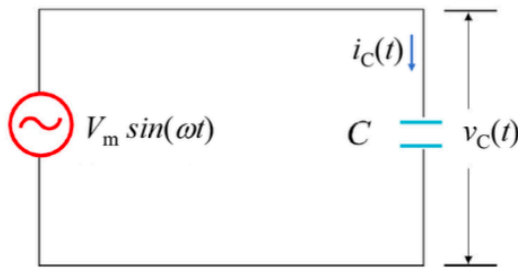
The product ωL is called the **Reactance** of L , where $X_L = \omega L$ (Recall: $\omega = 2\pi f$)

$$I_m = \frac{V_m}{\omega L} = \frac{V_m}{X_L} \quad \& \quad I_{rms} = \frac{V_m}{\omega L} = \frac{V_{rms}}{X_L}$$

$$\phi = -\frac{\pi}{2} \text{rad or } -90^\circ \text{ (Current lags Voltage)}$$

The **reactance of an inductor** is the *ratio between the voltage amplitude and the current amplitude*. The SI unit of reactance X_L is **ohm** (Ω). Unlike R , the value of X_L varies with frequency. For any given value of L , the resistance to current flow increases with increasing frequency of the source voltage.

Capacitor



$$v_C(t) = V_m \sin(\omega t)$$

$$i_C(t) = C \frac{dv_C}{dt}$$

$$i_C = \omega C V_m \sin(\omega t + \frac{\pi}{2}) = I_m \sin(\omega t + \frac{\pi}{2})$$

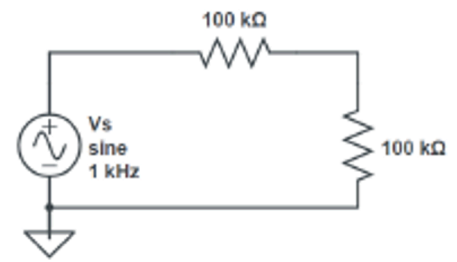
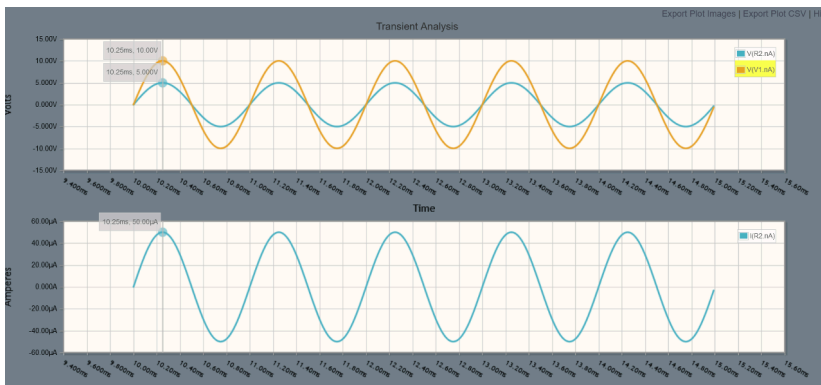
The reciprocal of the product ωC is called the **reactance** of C , where $X_C = \frac{1}{\omega C}$

$$I_m = \omega C V_m = \frac{V_m}{X_C} \quad \& \quad I_{rms} = \omega C V_{rms} = \frac{V_{rms}}{X_C}$$

$$\phi = +\frac{\pi}{2} \text{rad or } +90^\circ \text{ (Current leads Voltage)}$$

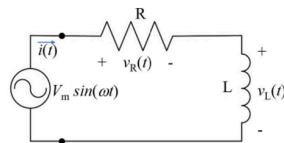
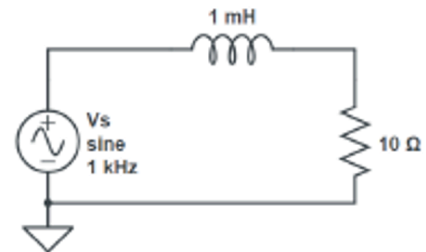
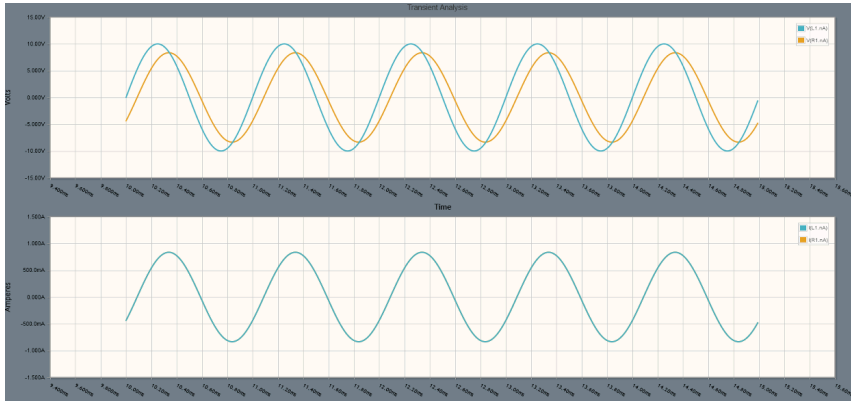
The SI units of X_C is ohm (Ω). The value of X_C is inversely proportional to ω . Thus, for a capacitor, the resistance to current flow decreases with increasing frequency of the source voltage.

R-R



L-R

From analysis, current is **lagging** voltage, but phase shift is not equal to $\pi/2$



Applying KVL to the circuit:

$$v_R(t) + v_L(t) = V_m \sin(\omega t)$$

It's known:

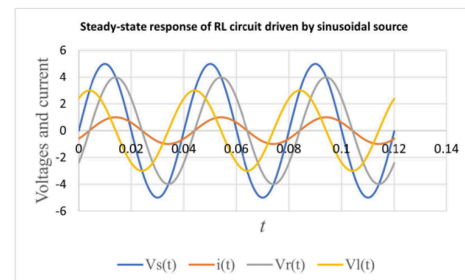
$$v_R = Ri \quad \text{and} \quad v_L = L \frac{di}{dt}$$

So that,

$$Ri + L \frac{di}{dt} = V_m \sin(\omega t)$$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{V_m}{L} \sin(\omega t)$$

Again, we illustrate the result by the steady-state plotting below:



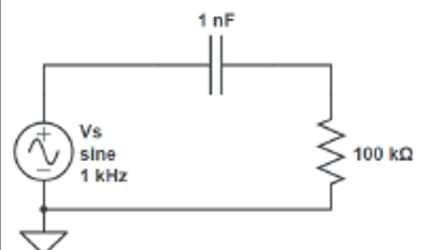
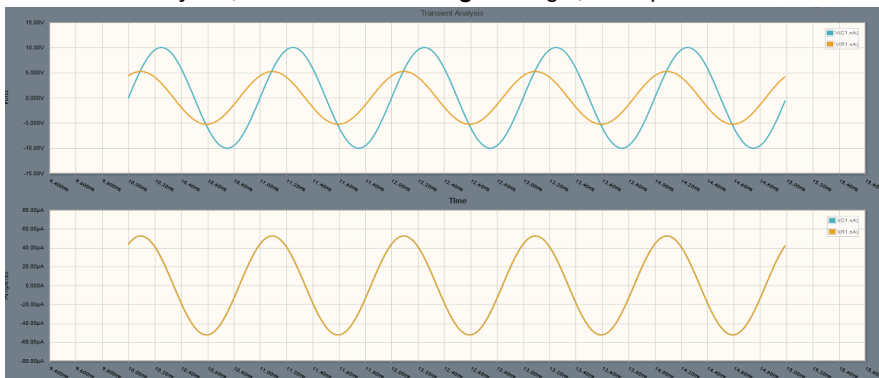
Some observations:

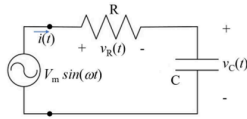
- Frequency is same for all voltages and current
- $i(t)$ is in-phase with $V_r(t)$
- $i(t)$ lags $V_L(t)$ by $\frac{\pi}{2}$ rad **Current lags voltage by $(\pi/2)$**
- $i(t)$ lags $V_S(t)$ but the phase shift is less than $\frac{\pi}{2}$ rad

Due to resistor

C-R

From the analysis, current is **leading** voltage, but phase shift is not equal to $\pi/2$





Applying KVL to the circuit:

$$v_R(t) + v_C(t) = V_m \sin(\omega t)$$

It's known:

$$v_R = Ri = RC \frac{dv_C}{dt} \quad i_C(t) = C \frac{dv_C}{dt}$$

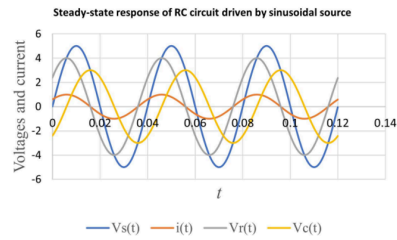
So that,

$$RC \frac{dv_C}{dt} + v_C(t) = V_m \sin(\omega t)$$

$$\frac{dv_C}{dt} + \frac{1}{RC} v_C(t) = \frac{V_m}{RC} \sin(\omega t)$$

The solution of this non-homogeneous first-order ordinary differential equation (ODE) is the capacitor voltage $v_C(t)$. Once $v_C(t)$ is obtained, we can determine the current and voltage.

Solving ODE is outside the scope of this course. We will illustrate the result by the steady-state plotting below:

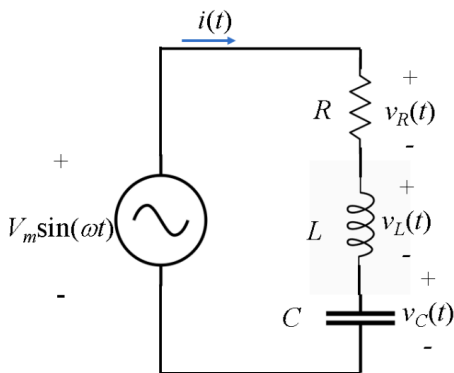


Some observations in the steady state:

- Frequency is same for all voltages and current
- $i(t)$ is in-phase with $V_r(t)$
- $i(t)$ leads $V_C(t)$ by $\frac{\pi}{2}$ rad **Current leads voltage by $(\pi/2)$**
- $i(t)$ leads $V_S(t)$ but the phase shift is less than $\frac{\pi}{2}$ rad

Due to resistor

R-L-C



$$V_{s,rms}^2 = V_{R,rms}^2 + (V_{L,rms} - V_{C,rms})^2$$

In an RLC circuit, resonance occurs when the source frequency of the AC supply matches the natural frequency of the circuit.

Let the current in a series RLC be

$$i(t) = I_m \sin(\omega t)$$

Then the voltage across the L and the C, respectively, are

$$v_L(t) = \omega L I_m \sin(\omega t + \frac{\pi}{2})$$

$$v_C(t) = \frac{I_m}{\omega C} \sin(\omega t - \frac{\pi}{2})$$

v_C and v_L are π rad out of phase.

At a particular frequency (ω_r), the amplitude of v_C and v_L are equal:

$$\omega_r L I_m = \frac{I_m}{\omega_r C} \rightarrow \omega_r = \frac{1}{\sqrt{LC}}$$

Resonant Frequency of series RLC in rad/s

Where ω_r is the natural frequency of the circuit where the circuit resonances.

At resonance, the amplitude of v_C is equal to the amplitude of v_L , and they are π rad out of phase. So, $v_C(t) + v_L(t) = 0$ and the voltage across the R is

$$v_R(t) = v_S(t) - (v_C(t) + v_L(t)) = v_S(t)$$

So, for a given amplitude of the source voltage, the amplitude of the current is **maximum** at the resonant frequency.

Resonance

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{1}{R} \times \sqrt{L/C}$$

- At resonance frequency:
 - Phase shift = 0
 - *Current Amplitude maximized*
 - The RLC circuit behaves like a **pure resistive circuit**

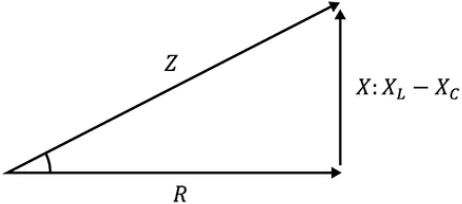
- Source frequency lower than resonance frequency
 - Phase Shift > 0 , **current leading source voltage for less than $\pi/2$**
 - *Current Amplitude reduced*
 - The RLC Circuit behaves like an RC circuit
- Source frequency higher than resonance frequency:
 - Phase Shift < 0 , **current lagging source voltage for less than $\pi/2$**
 - *Current Amplitude reduced*
 - The RLC circuit behaves like an RL circuit

Impedance

Impedance, represented by the symbol Z , is **a measure of the opposition to electrical flow**. It is measured in ohms (Ω).

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Circuit element	Impedance	
	Symbol	Value
Resistor	R	R
Capacitor	X_C	$\frac{1}{j\omega C}$
Inductor	X_L	$j\omega L$



$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$